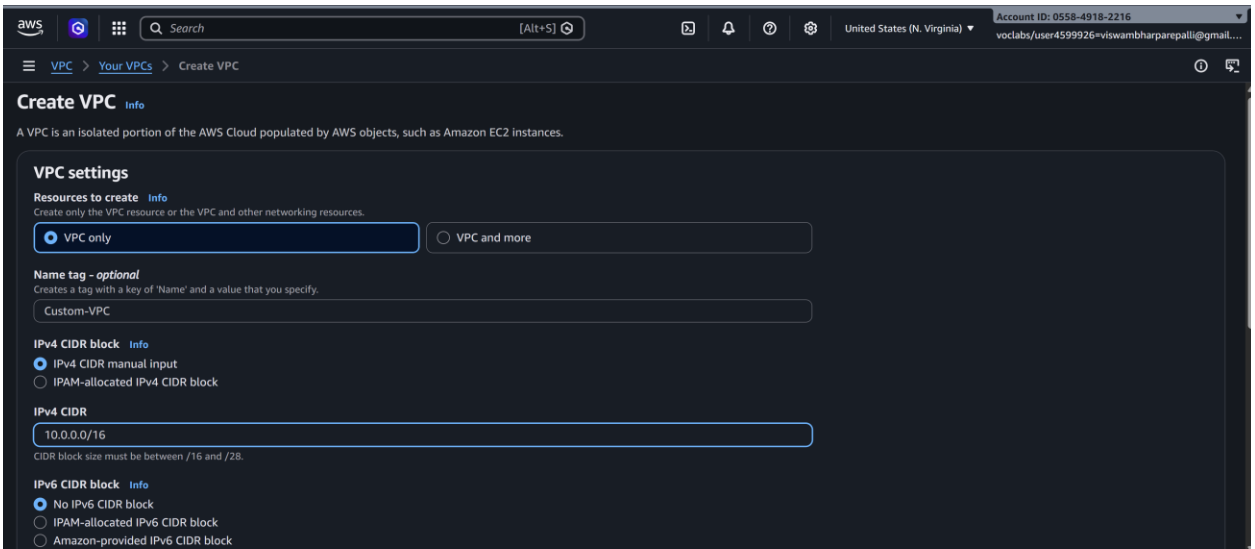
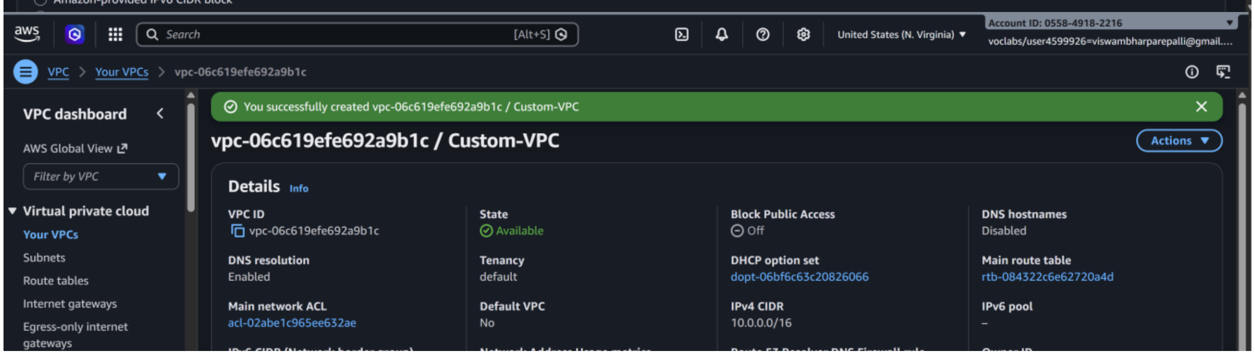
Week 5: VPC Creation and Ec2 Instance Connection

◆ Step 1: Create the VPC

1. Go to **AWS Management Console → VPC**
2. Click **Create VPC**
3. Choose **VPC only**
4. Enter:
    - **Name**: `Custom-VPC`
    - **IPv4 CIDR block**: e.g. `10.0.0.0/16`
5. Click **Create VPC**
6. 

7. 

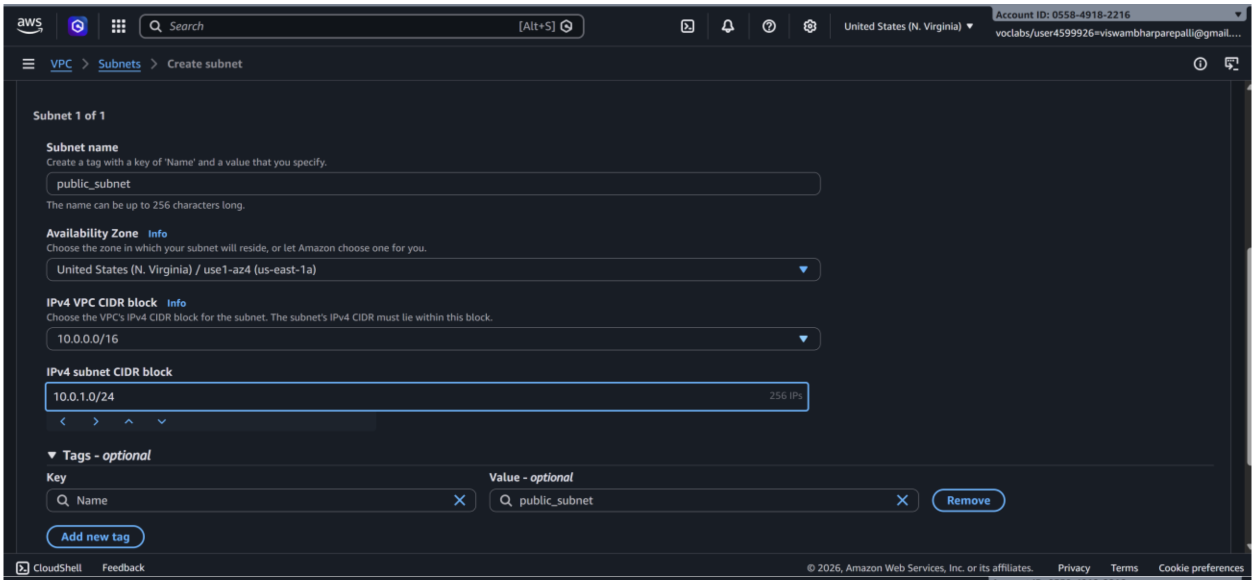
◆ Step 2: Create Subnets (Public & Private)

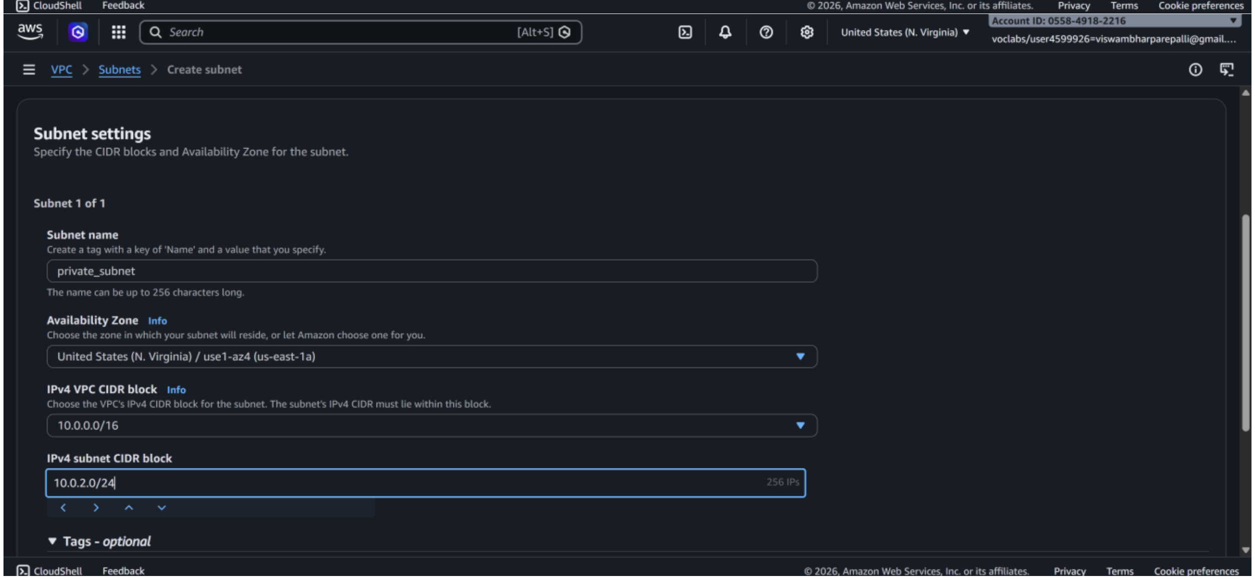
Create subnets in same **Availability Zones** for high availability.

Example:

- **Public Subnet**: `10.0.1.0/24` (AZ-A)
- **Private Subnet**: (AZ-A)

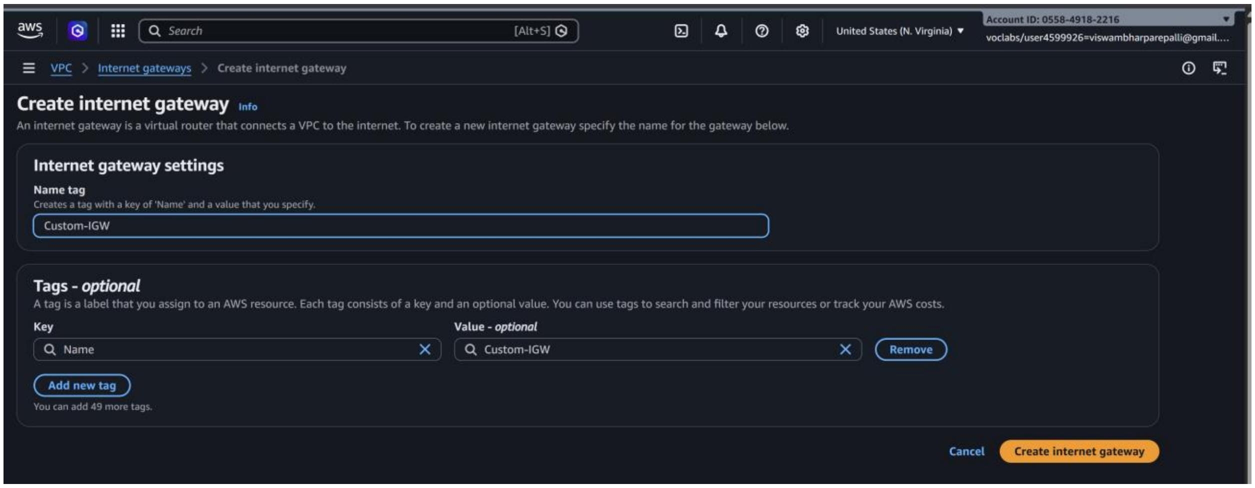
Steps:

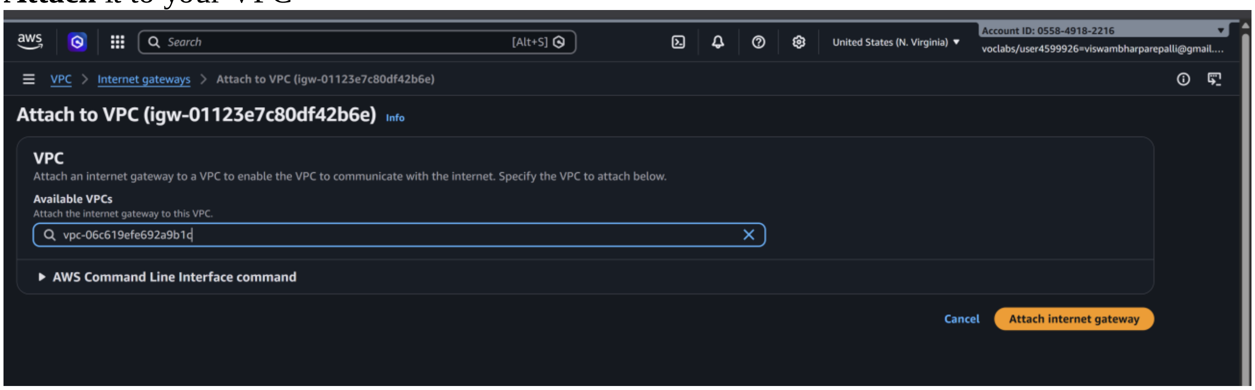
1. Go to **Subnets → Create subnet**
2. Select your VPC
3. Choose AZ
4. Enter CIDR
5. Create subnet
6. 

7. 

◆ Step 3: Create an Internet Gateway (IGW)

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name it `Custom-IGW`

4. 

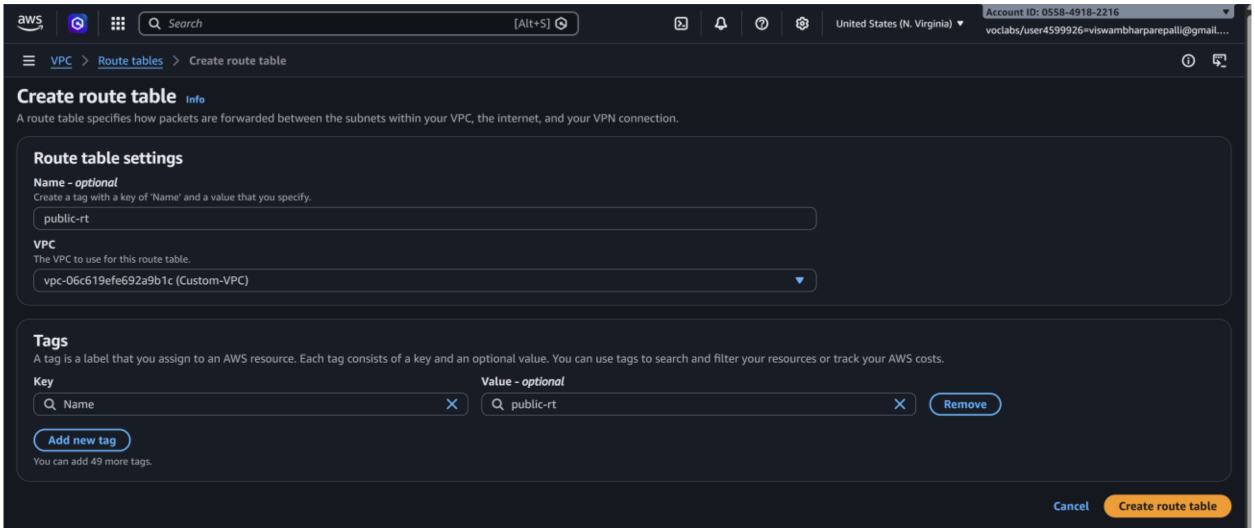
5. **Attach** it to your VPC
6. 

◆ Step 4: Create Route Tables

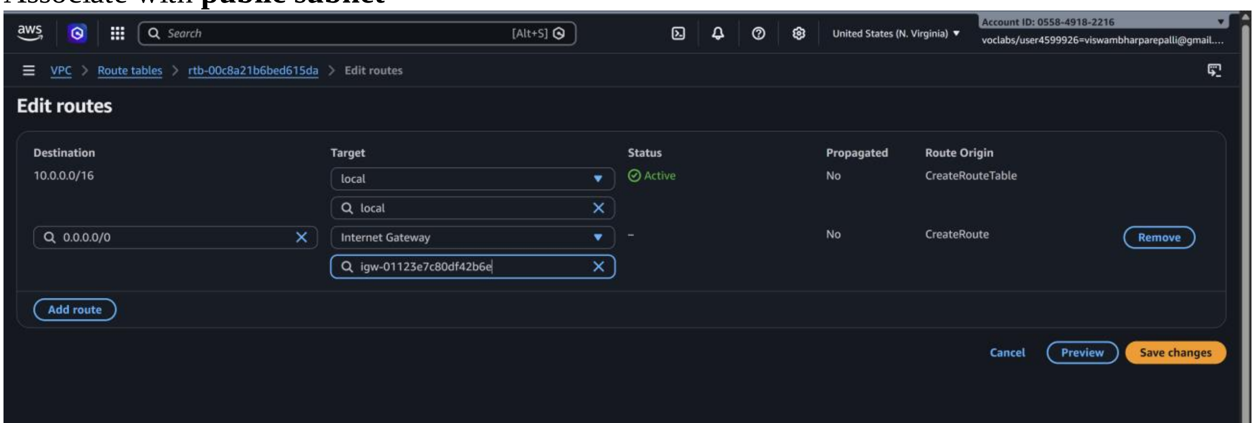
You need **separate route tables** for public and private subnets.

Public Route Table

1. Go to **Route Tables → Create route table**
2. Select VPC



3.
4. Add route:
   - Destination: `0.0.0.0/0`
   - Target: **Internet Gateway**
5. Associate with **public subnet**



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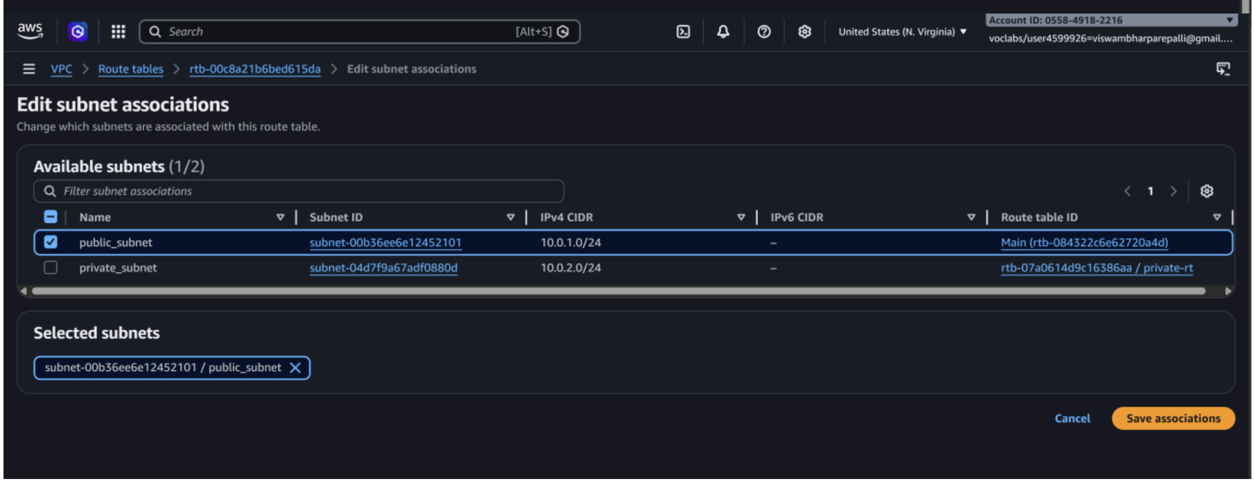
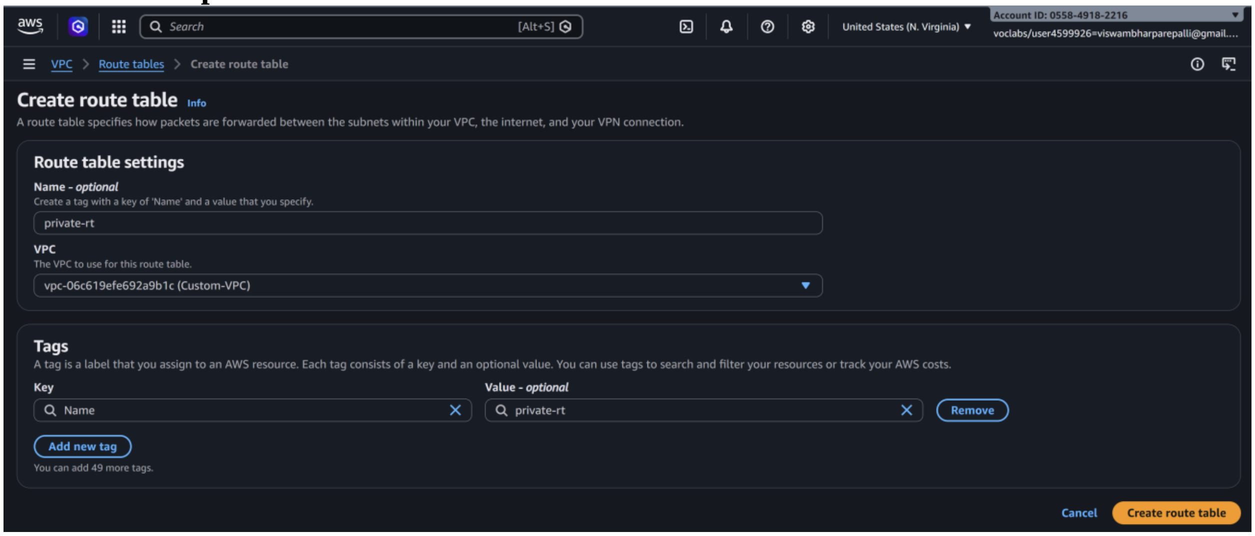


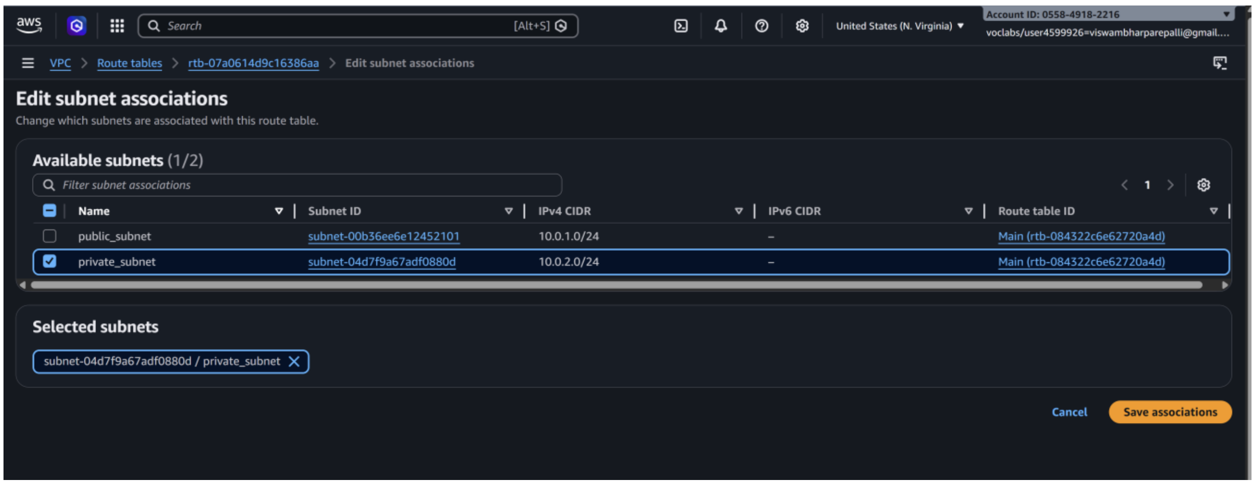
7.

Private Route Table

- Keep default local route only (no IGW)

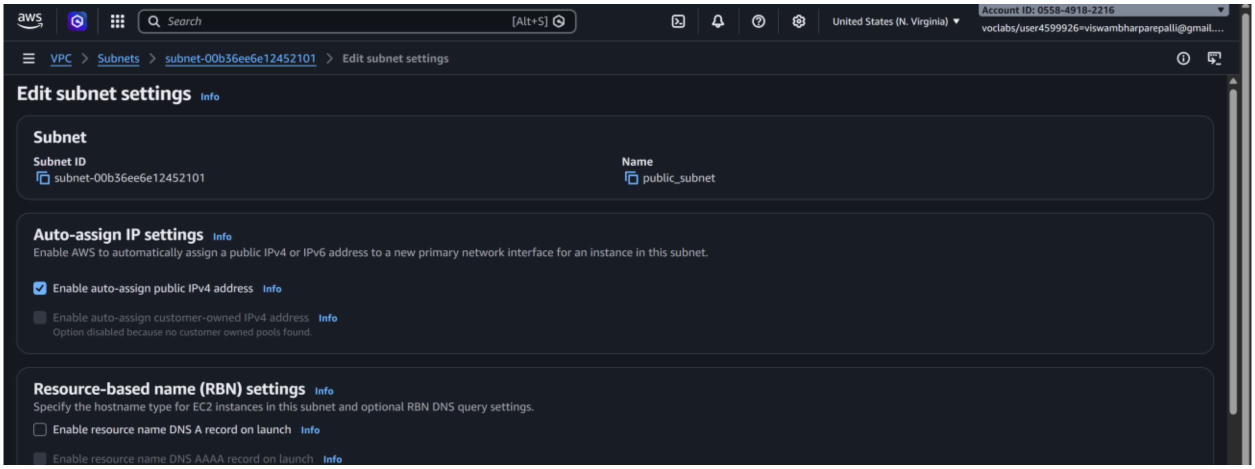
- Associate with **private subnet**



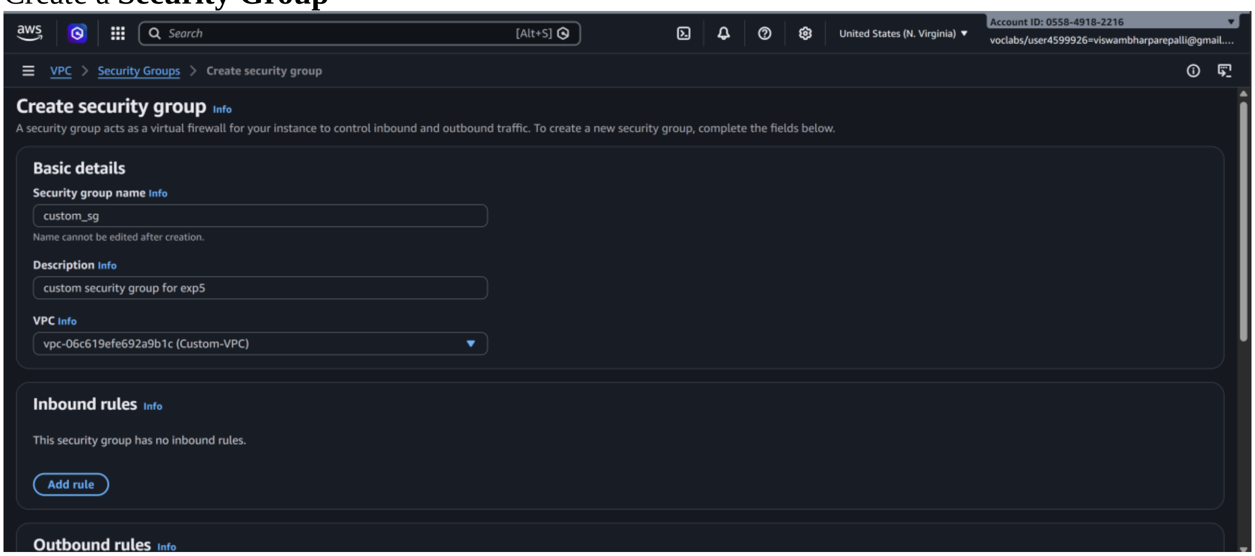
- 

◆ Step 5: Enable Auto-Assign Public IP (Public Subnet)

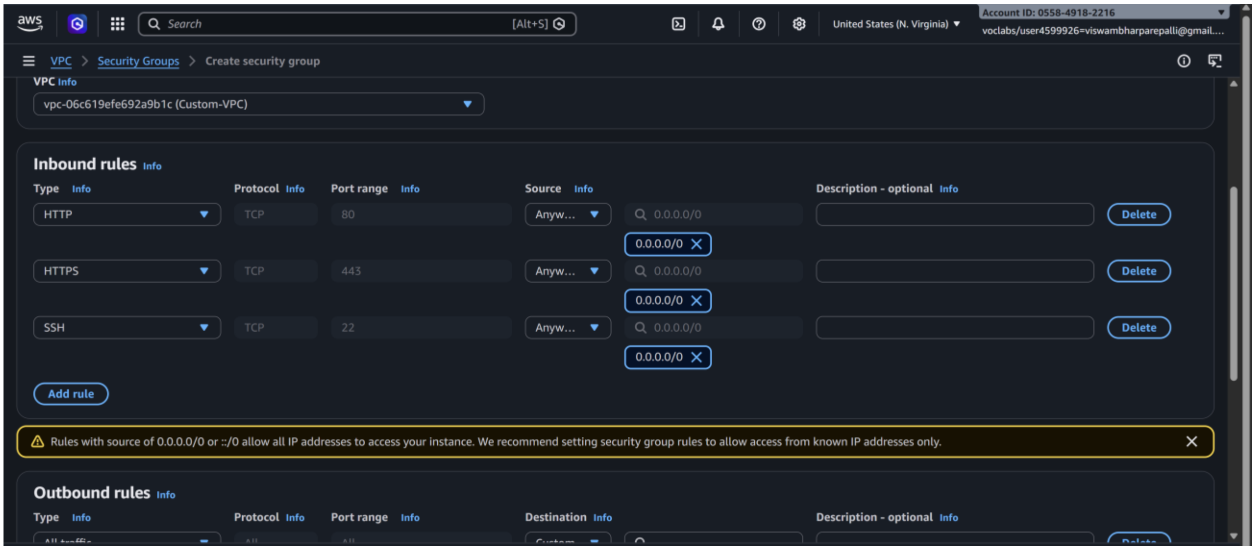
1. Select **Public Subnet**
2. Go to **Edit subnet settings**
3. Enable **Auto-assign public IPv4 address**

4. 

◆ Step 6: Configure Security Groups

1. Create a **Security Group**
2. 

3. Add inbound rules:
   - HTTP (80) / HTTPS (443)
   - SSH (22) from trusted IP



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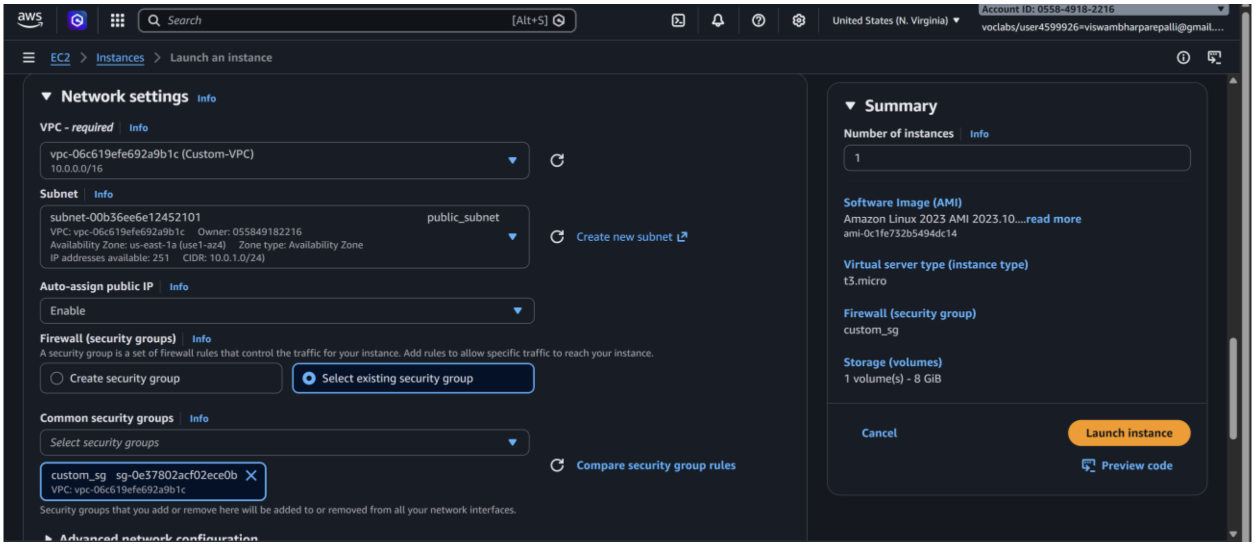
4. Attach to EC2 instances

◆ Step 7: Launch EC2 Instances

- Public EC2 → Public subnet



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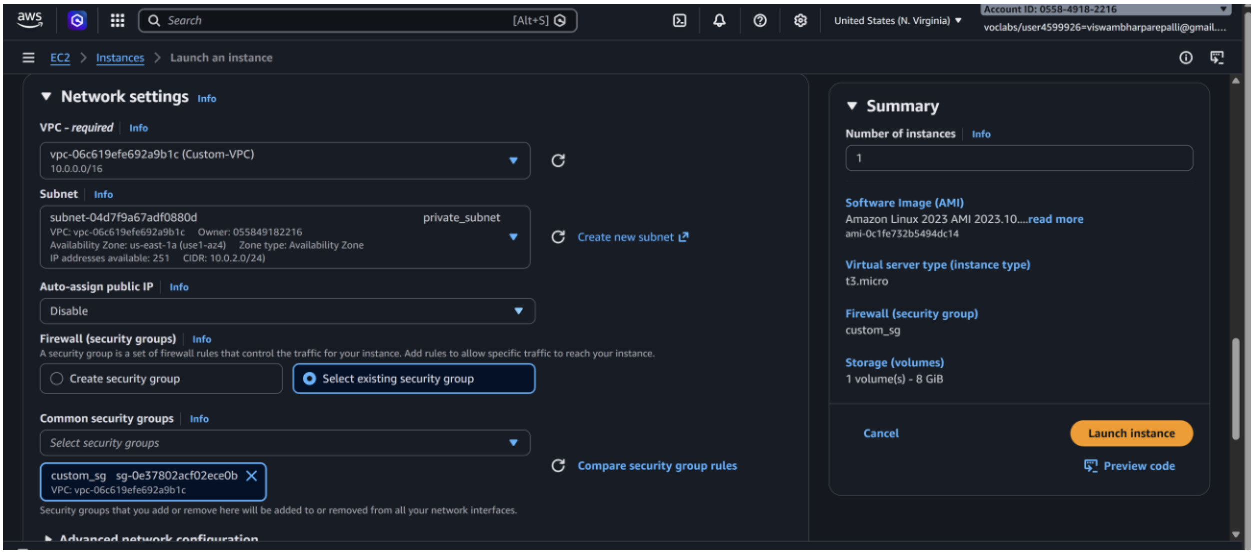
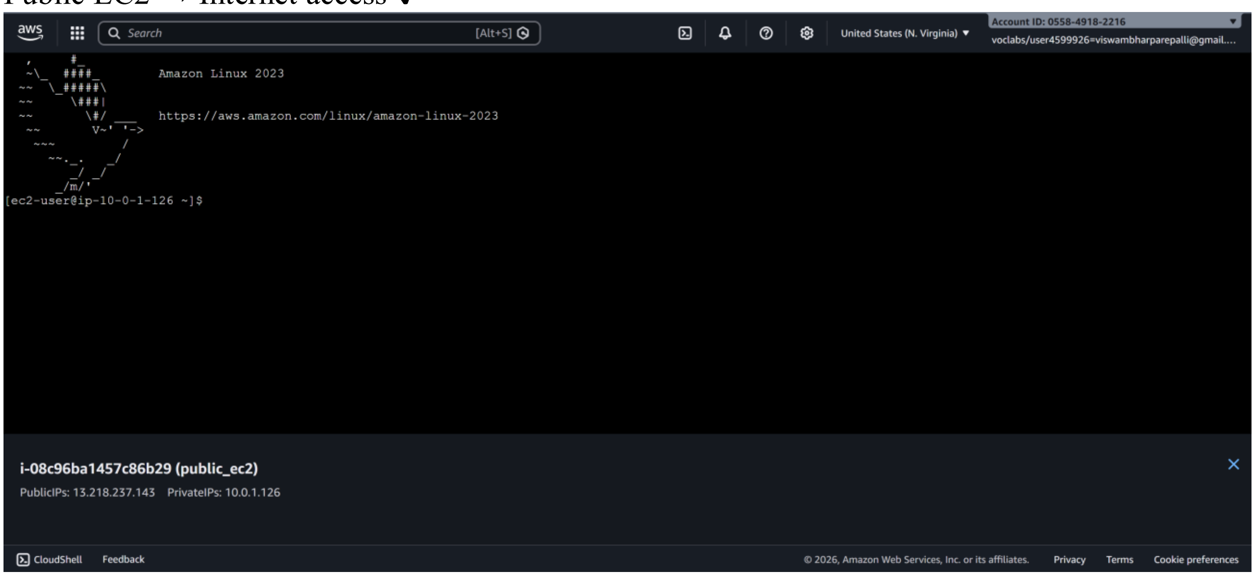
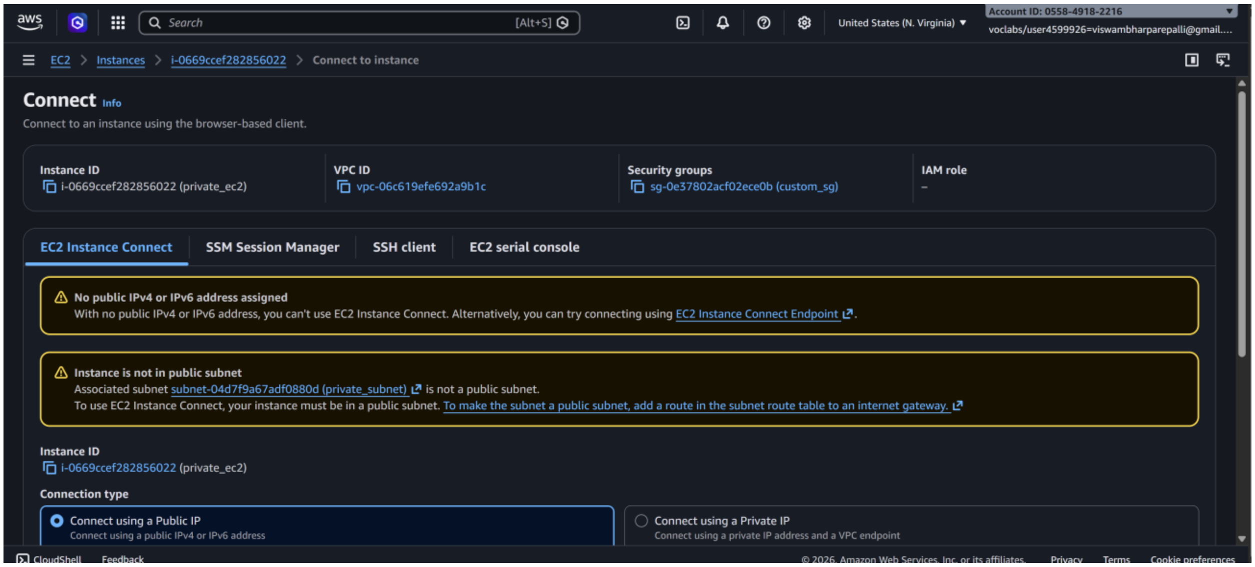
- Private EC2 → Private subnet
  Attach correct security groups.

- 

◆ Step 8: Test the Setup

- Public EC2 → Internet access ✓
- 

- Private EC2 → Internet via NAT ✓
- Private EC2 → No direct inbound access ✓



Scenario Based Questions:

1. You have a web application where users access a website, but the database should not be exposed to the internet. How would you design the VPC? Which resources go into public and private subnets?
   Answer:
   Use a layered subnet design:

- Public subnets: Internet-facing Application Load Balancer, NAT Gateway, Bastion host (if needed).
- Private subnets: Web/App servers and RDS database.
  Attach an Internet Gateway (IGW) to the VPC. Public subnets route 0.0.0.0/0 → IGW. Private subnets use NAT Gateway for outbound internet. Database stays isolated with no public IP.

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2. An EC2 instance in a public subnet cannot access the internet. What VPC components would you check and why?
   Answer:
   Check in this order:

- Internet Gateway attached to VPC.

- Route table: 0.0.0.0/0 → IGW.
- Instance has public IP or Elastic IP.
- Security group outbound rules.
- NACL outbound rules.
- Source/Destination check (if NAT/bastion).
  Most failures are missing public IP or wrong route table.

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3. Your application server needs to connect to an RDS database securely. How would you configure security groups and subnets?
   Answer:

- Place RDS in private subnets (no public access).
- App servers in private or public subnets depending on architecture.
- Create separate security groups:
  - App SG: allow outbound DB port.
  - RDS SG: allow inbound DB port only from App SG (not CIDR).
    This enforces least privilege and avoids exposing DB publicly.

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4. You are asked to design a VPC for 500 servers today, but it should scale to 2,000 servers in the future. How would you choose the CIDR block?
   Answer:
   Plan for future growth:

- 2,000 instances require far more IPs due to scaling buffers, ALBs, RDS, Lambda ENIs, etc.
- Choose something like /16 CIDR (e.g., 10.0.0.0/16) instead of /24 or /20.
  AWS reserves 5 IPs per subnet, so larger CIDR prevents painful re-architecture later.

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5. A company wants secure connectivity between its on-premises data center and AWS VPC. Which AWS services would you choose and why (VPN vs Direct Connect)?
   Answer:

- Site-to-Site VPN: Fast to deploy, encrypted over internet, lower cost. Good for quick setup or moderate traffic.

- Direct Connect: Dedicated private link, predictable latency, high bandwidth, no internet exposure. Best for production hybrid workloads.
  Often companies use Direct Connect + VPN backup.

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6. You need to connect two VPCs, but both use the same CIDR range. How would you solve this problem?
   Answer:
   VPC peering and Transit Gateway don't support overlapping CIDRs. Options:

- Re-IP one VPC (best long-term solution).
- Use NAT Gateway or proxy-based routing.
- Use PrivateLink if only specific services need exposure.
  Overlapping CIDRs are a design mistake — avoid them early.

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7. Multiple microservices running in different subnets need to communicate securely. How would you design routing and security groups?
   Answer:

- Keep default local route in route tables for intra-VPC communication.
- Use security-group-to-security-group rules, not wide CIDR rules.
- Separate SG per service (e.g., ServiceA SG allows inbound only from ServiceB SG).
- Avoid opening entire subnet ranges unless absolutely necessary.

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8. Design a VPC for a 3-tier application (Web, App, DB) with high security and scalability. Explain subnets, route tables, gateways, and security groups.
   Answer:
   Architecture:

- Public subnets: ALB, NAT Gateway.
- Private App subnets: Application servers.
- Private DB subnets: RDS.

Routing:

- Public route table → IGW.
- App subnet route → NAT Gateway.
- DB subnet → no internet route.

Security Groups:

- ALB SG allows 80/443 from internet.
- App SG allows traffic only from ALB SG.
- DB SG allows DB port only from App SG.

Use multi-AZ subnets for high availability.

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9. You suspect unusual traffic inside your VPC. How would you monitor and analyze network traffic?
   Answer:

- Enable VPC Flow Logs → CloudWatch or S3.
- Use CloudWatch metrics and alarms.
- AWS GuardDuty for threat detection.
- Traffic Mirroring for deep packet inspection if needed.
  Flow logs are usually the first step.

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10. Why can't a private subnet have an Internet Gateway directly attached?
    Answer:
    Internet Gateways attach to the VPC, not individual subnets.
    A subnet becomes "public" only when its route table sends traffic to the IGW. Private subnets intentionally avoid that route for security isolation.

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11. What happens if route tables are misconfigured in a VPC?
    Answer:
    Traffic gets dropped or blackholed:

- Instances lose internet access.
- Services cannot reach each other.
- Load balancers fail health checks.
  Routing issues often look like application failures but are network-layer problems.

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12. What happens if a route table has no local route?
    Answer:
    In AWS you cannot remove the local route — it is automatically added.

If hypothetically missing, resources inside the same VPC couldn't communicate across subnets.

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13. An EC2 instance allows traffic on port 80 in the security group, but traffic is still blocked. What could be the reason?
    Answer:
    Common causes:

- NACL blocking inbound/outbound.
- Wrong route table.
- Instance has no public IP.
- Web server not running or listening.
- OS firewall (iptables/Windows firewall).
  Security group alone doesn't guarantee connectivity.

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14. You need to create a VPC that will host 1,000+ EC2 instances across multiple AZs. How do you decide the CIDR block?
    Answer:
    Estimate future IP usage, not just EC2 count:

- Include scaling buffers, load balancers, RDS, endpoints.
- Use /16 or /17 CIDR to avoid fragmentation.
- Pre-plan subnet ranges per AZ (e.g., /20 per tier per AZ).

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15. Only a company's corporate IP should be able to SSH into EC2 instances. How would you implement this securely in AWS VPC?
    Answer:

- In EC2 security group inbound rule:
  - Allow TCP 22 from corporate public IP CIDR (e.g., 203.x.x.x/32).
- Prefer bastion host or AWS Systems Manager Session Manager to avoid exposing SSH directly.

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16. An EC2 instance can send traffic out but cannot receive responses. Which VPC component might be misconfigured and why?

Answer:
Likely issues:

- Network ACL blocking inbound ephemeral ports (stateless).
- Asymmetric routing.
- Security group missing inbound rule for return traffic (less common because SGs are stateful).
- NAT Gateway/instance misconfiguration.
NACLs are the usual culprit because they require explicit inbound and outbound rules.